column-hse-ltau: hydrostatic columns on an optical-depth grid

The physics of `run.mode = "column-hse-ltau"`, explained step by step (B.Sc. level). HSEMHS.

1. The question

When we observe the Sun we do not see heights; we see light. A Stokes inversion therefore returns the temperature and the magnetic field not at fixed geometric heights but on surfaces of constant **optical depth** τ_{500}: at $\log_{10}\tau_{500} = 0, -1, -2, \ldots$ in every pixel. To turn such an atmosphere into a physical one we need, in every pixel, the gas pressure and the density on each τ surface, and the **geometric height** at which that surface lies. This mode computes them by assuming that every pixel column is in hydrostatic equilibrium on its own.

2. The quantities

Symbol	Meaning	Unit
τ	optical depth at 500 nm, counted from the top downward; the grid levels are equally spaced in $\log_{10}\tau$	–
T, $\mathbf{B}$	temperature and magnetic field on the τ levels, given, never changed	K, T
P, ρ	gas pressure and density on the τ levels	Pa, $\mathrm{kg\,m^{-3}}$
χ	extinction coefficient at 500 nm (fraction of the light removed per metre)	$\mathrm{m^{-1}}$
κ	extinction per unit mass, $\kappa = \chi/\rho$	$\mathrm{m^2\,kg^{-1}}$
m	column mass: mass of gas above a point per unit area	$\mathrm{kg\,m^{-2}}$
$z(\tau)$	geometric height of a τ level in a column	m
g	solar surface gravity, 274	$\mathrm{m\,s^{-2}}$
μ_0	vacuum permeability, $4\pi \times 10^{-7}$	$\mathrm{T\,m\,A^{-1}}$

3. Optical depth is a measure of mass

Optical depth is defined by how much light is removed on the way out: going down a distance dz adds $d\tau = \chi\,dz$ (with z upward, $d\tau = -\chi\,dz$). Write $\chi = \kappa\rho$: the extinction is the product of how much gas there is and how strongly each kilogram of it absorbs. Then

$$d\tau = -\kappa\,\rho\,dz = \kappa\,dm,$$

because $-\rho\,dz = dm$ is the mass added to the column above as we go down by dz. So optical depth is *column mass weighted by the opacity*. For a gas with constant κ, $\tau = \kappa\,m$ exactly.

4. Hydrostatic equilibrium in optical depth

On a height grid, hydrostatic equilibrium reads $dP/dz = -\rho g$ (the weight of a thin slab is carried by the pressure difference across it). Divide it by $d\tau = -\kappa\rho\,dz$:

$$\boxed{\frac{dP}{d\tau} = \frac{g}{\kappa}}$$

This is the classical form used in stellar atmospheres. Since $dP = g\,dm$ (the pressure is the weight of the gas above), it simply says that adding the optical depth $d\tau$ means adding the mass $d\tau/\kappa$. With constant κ it integrates to $P = g\tau/\kappa$: *the pressure at optical depth τ is the weight of the gas that produces that much opacity.*

The levels are equally spaced in $\log_{10}\tau$, so the code integrates in $\ln\tau$ ($d\tau = \tau\,d\ln\tau$) and in $\ln P$ ($dP = P\,d\ln P$):

$$\boxed{\frac{d\ln P}{d\ln\tau} = \frac{g\,\tau}{\kappa\,P} = \frac{g\,\tau\,\rho}{\chi\,P}}$$

ρ and χ depend on T and P through the equation of state (ideal gas with ionisation and molecules, from Wittmann's code, tabulated in $(\ln T, \ln P)$). The equation is therefore integrated numerically, level by level from the top down.

5. Why no externally supplied reference pressure is needed

On a height grid the pressure needs a prescribed value somewhere (an anchor), because a height coordinate says nothing about how much gas there is: the same temperature profile is equally hydrostatic at any pressure level. Optical depth is different: τ measures the mass above a point, and hydrostatic equilibrium says that the pressure is the weight of that mass. Once $T(\tau)$ and the opacity of the gas are known, the pressure follows, and no reference value has to be supplied from outside.

This does not mean the result is free of assumptions. Three modelling choices enter instead: the closure at the top of the column (next section), the treatment of sharp temperature jumps (Sect. 8) and, for the heights, the alignment of the columns (Sect. 9). And the result is only as good as the opacity: if the τ scale of the input was computed with a different opacity, the pressures differ in proportion.

6. The top of the column

The integration needs a starting value at the top level τ_1 (here 10^{-8}). Above it we assume κ stays constant at its value κ_1 and that the pressure vanishes at $\tau = 0$. Then the equation integrates to

$$P_1 = \frac{g\,\tau_1}{\kappa_1(T_1, P_1)}.$$

Because κ_1 itself depends on P_1, this is solved by iteration (guess P_1, compute κ_1, update P_1, repeat). An error in this assumption matters only near the top: deeper down the pressure is dominated by the gas added below τ_1. Roughly, its relative weight is τ_1/τ, i.e. 10^{-8} at $\tau = 1$.

7. Heights from the opacity

A τ level has a height only once we know how thick each layer is. From $d\tau = -\chi\,dz$:

$$\boxed{\frac{dz}{d\ln\tau} = -\frac{\tau}{\chi}}$$

Where the gas is opaque (large χ), a step in $\ln\tau$ is a thin layer; where it is transparent, the same step is a thick layer. z is integrated alongside $\ln P$. This fixes the heights in each column only *up to an additive constant*: we know the distance between the levels, not where the column sits (Sect. 9).

8. Between grid levels: hot gas over cool gas

The grid has 10 levels per factor of ten in τ. Between two levels $\ln T$ is assumed to vary linearly with $\ln\tau$, and the integration takes small steps so that $\ln P$ changes by at most 0.05 per step.

The opacity at 500 nm comes mainly from the negative hydrogen ion H^-, which needs free electrons, and both rise very steeply with temperature. So cool gas is far more transparent than hot gas. Suppose a hot layer (8000 K) lies directly on a cool one (4000 K) within one grid interval. With the linear interpolation, a whole tenth of a decade in τ is then filled with nearly transparent gas. By $dP/d\tau = g/\kappa$, collecting that much optical depth from transparent gas requires a huge amount of mass, and the pressure jumps by orders of magnitude within the interval. We regard this as mainly a consequence of representing a sharp temperature change by smooth interpolation on a coarse grid, though the true structure between two grid levels is not known. The code therefore limits the rise of $\ln P$ to 5 per 0.1 dex in τ (`cap_per_dex` = 50 per dex), a limit well above the rises of ordinary stratifications.

Where the limit acts, the column is *not* hydrostatic in that interval: the pressure is set to the limited value, and the integration of that interval stops there, so the rest of its geometric thickness is not added to the heights below (a known shortcoming of the present code). In the intervals below, the hydrostatic integration continues from the limited pressure; whether the pressure then comes back to what an exact integration would give depends on the column. The number of limited intervals is reported with every run, and results in those columns should be treated with care.

9. Where the columns sit: total-pressure alignment

9.1 Magnetic pressure

In each column we now know P, ρ and the layer thicknesses, but not the height of the column as a whole. The mode fixes it with a physical argument that involves the magnetic field. The force a magnetic field exerts on the gas, the Lorentz force, can be rewritten with a vector identity as

$$\mathbf{J}\times\mathbf{B} = \frac{(\nabla\times\mathbf{B})\times\mathbf{B}}{\mu_0} = -\nabla\left(\frac{B^2}{2\mu_0}\right) + \frac{(\mathbf{B}\cdot\nabla)\mathbf{B}}{\mu_0}.$$

The first term acts like an additional pressure, the **magnetic pressure** $B^2/2\mu_0$; the second is a tension along curved field lines.

9.2 The alignment rule

If the field is nearly vertical and only weakly curved, its tension is small and the horizontal force balance between neighbouring columns reduces to equal *total* pressure, $P + B^2/2\mu_0$, at the same height (a higher total pressure

would push the neighbour aside). The mode uses this as an **approximate** alignment rule, applied at a single height and ignoring the tension: it takes as reference the median, over all columns, of the total pressure on the deepest τ level, and shifts every column vertically so that its total pressure reaches the reference at one and the same height (interpolating $\ln(P + B^2/2\mu_0)$ linearly between levels). A column whose deepest level has a total pressure below the reference is extended downward, assuming that $\ln(P + B^2/2\mu_0)$ continues to rise with the *gas* pressure scale height of that level, which is a further approximation. With `align = "none"` the columns are not aligned at all (each has $z = 0$ at $\tau = 1$).

9.3 Consequence: the Wilson depression

Under this rule a column with a strong field carries part of the load by magnetic pressure, so it needs less gas pressure at the reference height. Less gas means a lower density, a lower opacity and a longer path to reach $\tau = 1$: its $\tau = 1$ surface tends to lie deeper. This is the Wilson depression seen in sunspots near the limb. It is an expected tendency, not a guarantee: the outcome for a given column also depends on its temperature structure and on the approximations above.

9.4 The gauge

Finally all columns are shifted together so that the average height of $\tau = 1$ is zero. This is only a choice of origin, with no physical content.

10. What the result means

- Every column is hydrostatic along τ, apart from the intervals where the pressure rise was limited (Sect. 8): there the pressure is not the weight of the gas above, and the heights below miss part of a layer.
- The heights follow from the opacity; their absolute value comes from the alignment (an approximation) and the gauge (a convention), not from a measurement. The field enters only through the alignment, via its magnetic pressure.
- The field values are taken as given on the τ levels wherever those levels end up in space; nothing makes the field divergence-free at its new positions.
- The columns are not in horizontal force balance with each other, and the tension of the field is ignored. Adding both is the purpose of `mhs-poisson-ltau`, which starts from this result.